

A Matrix Multiplication Accelerator Based on Systolic Array and Tiling

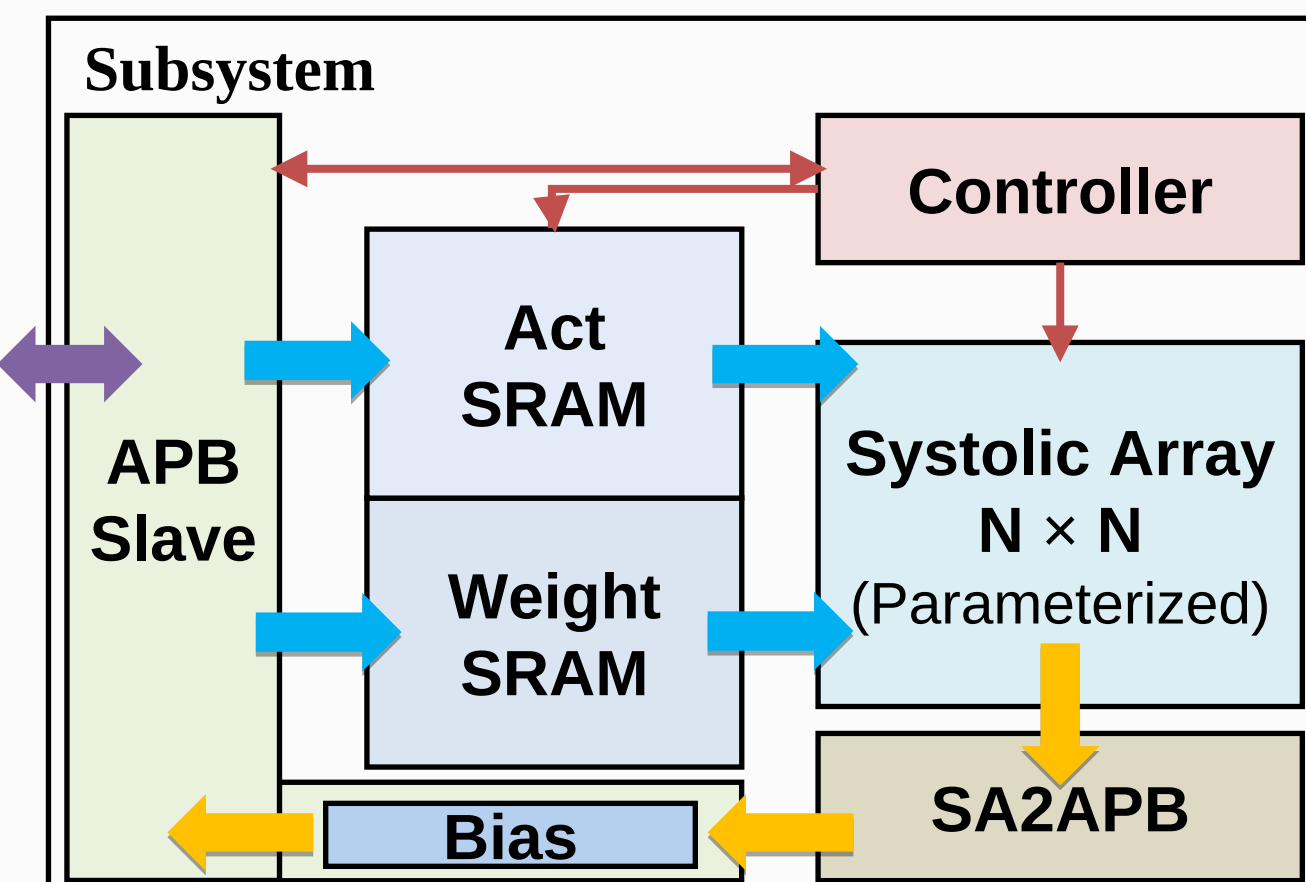


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Introduction

Traditional matrix multiplication is highly time-consuming, therefore, an accelerator based on a systolic array and a tiling strategy is proposed. The matrices are stored in a global SRAM, and an $N \times N$ systolic array tile is reused to compute the whole matrices. The subsystem is integrated into the Didactic SoC platform through an APB bus.



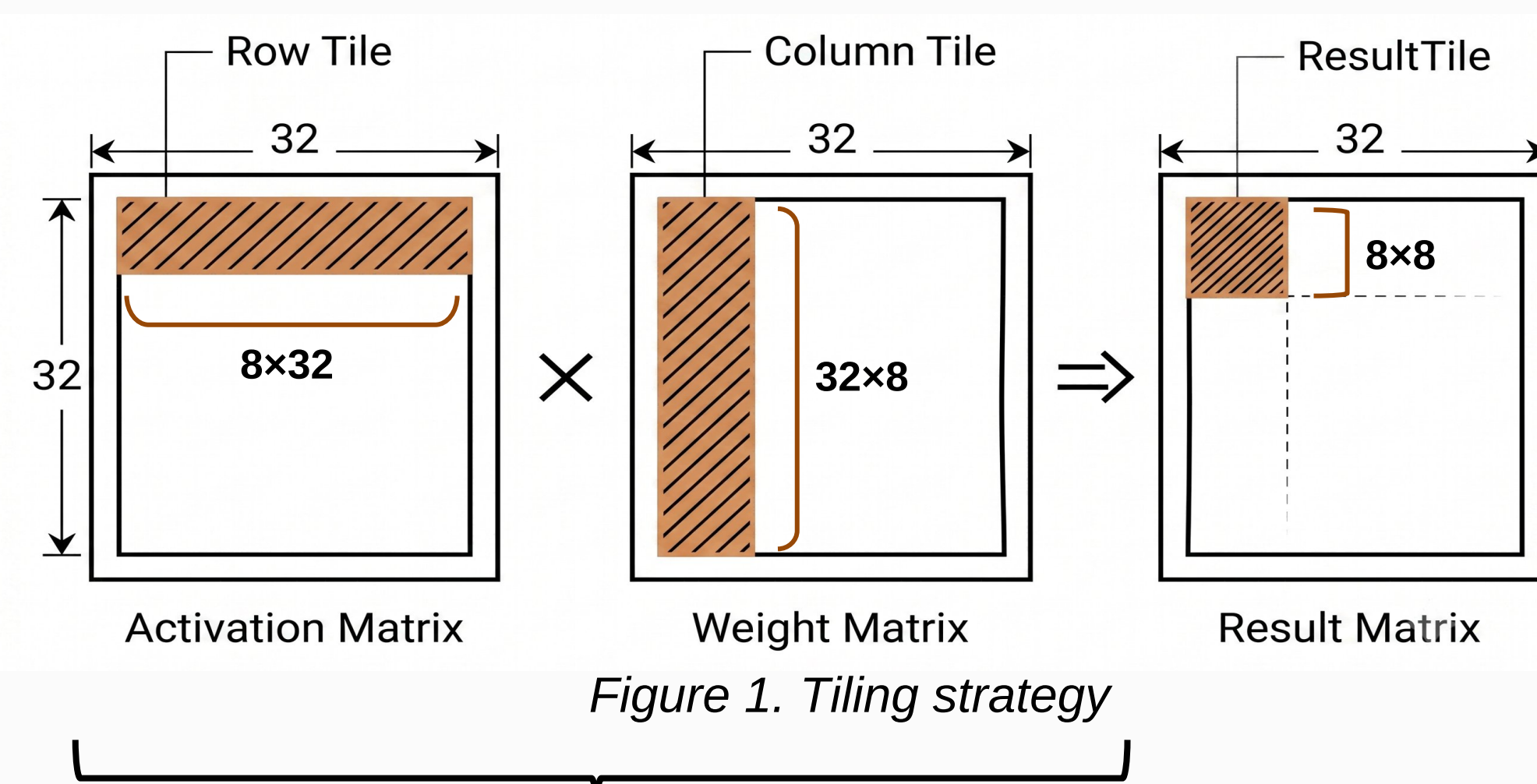
Highlights

- **Systolic Array:** Enables staggered parallel data flow into the MAC array and all units compute synchronously. The controller enables rows and columns' MAC units precisely.
- **Tiling:** Processes data iteratively in a small tile, dynamically adapting to hardware resources. Data is stored in a global SRAM, avoiding repeated low-speed APB transfers.
- **Adaptive MAC Unit:** Supports multi-precision signed integer computation from 8-bit to 32-bit. The internal multiplier can be configured as a combinational or multi-cycle Booth multiplier.

Implementation Details

Before Computation:

The Software Driver pre-processes the inputs and configures the accelerator according to the matrix data format. The data width of the activation and weight matrix is automatically detected, **supporting multiple precision modes**. Since the activation and weight matrices may use different data widths, the driver configures them separately. Also, to make the mix-precision data transfer more efficient, low-precision matrix elements are packed into 32bit before written.



Tiling Strategy (Example of N = 8):

Partition a 32×32 GEMM into 16 tiles of 8×8 . Rows are interleaved across 8 SRAM banks for conflict-free feeding. The tile is reused multiple times to calculate the big matrix. The tile size is parameterized, and 8×8 and 4×4 tiles were tested on an FPGA.

MAC Unit Design

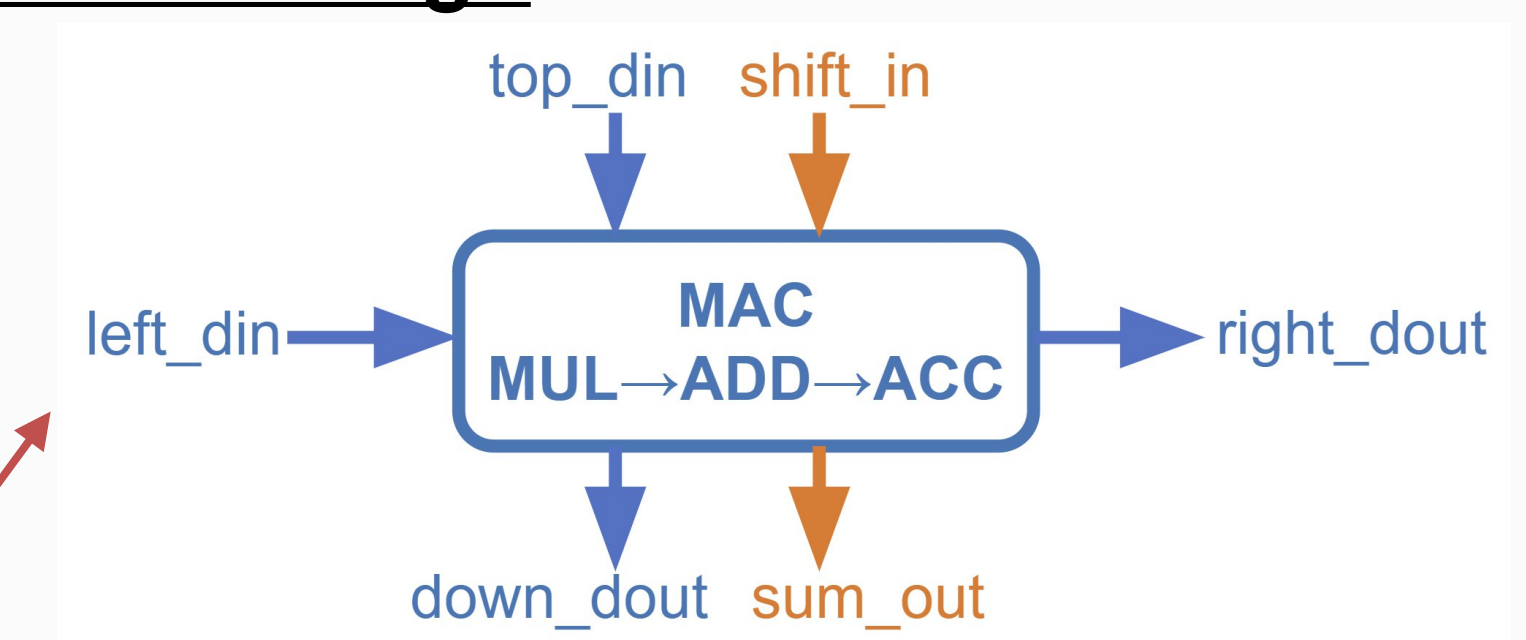


Figure 3. MAC Unit data ports

The **blue arrows** in Figure 3 are the SA calculation flow. The top and left data come from SRAM banks feeding into the MAC unit. The **orange arrows** are an accumulator shift chain used in data readout.

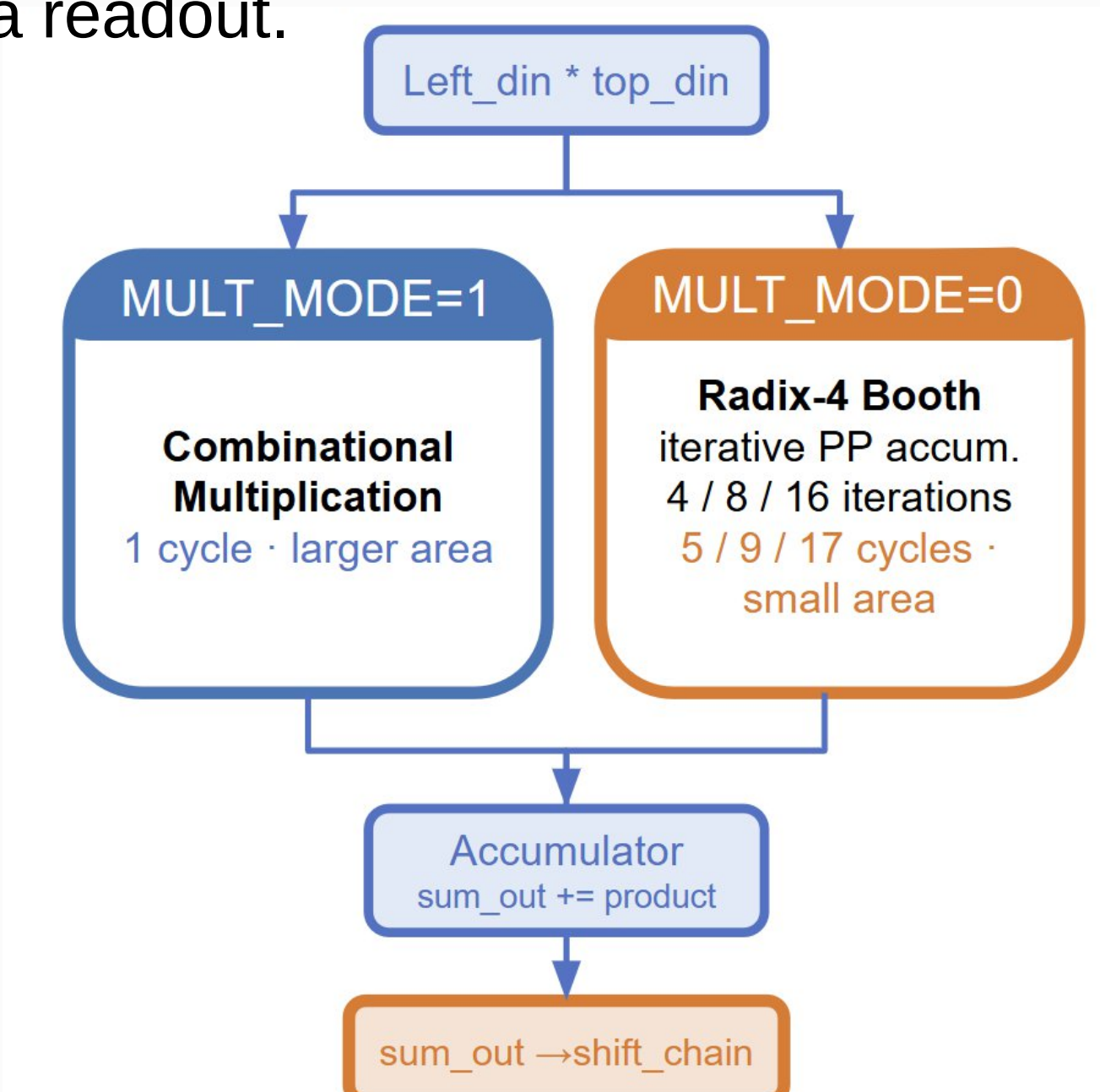


Figure 4. Parameterized Dual-Mode MAC

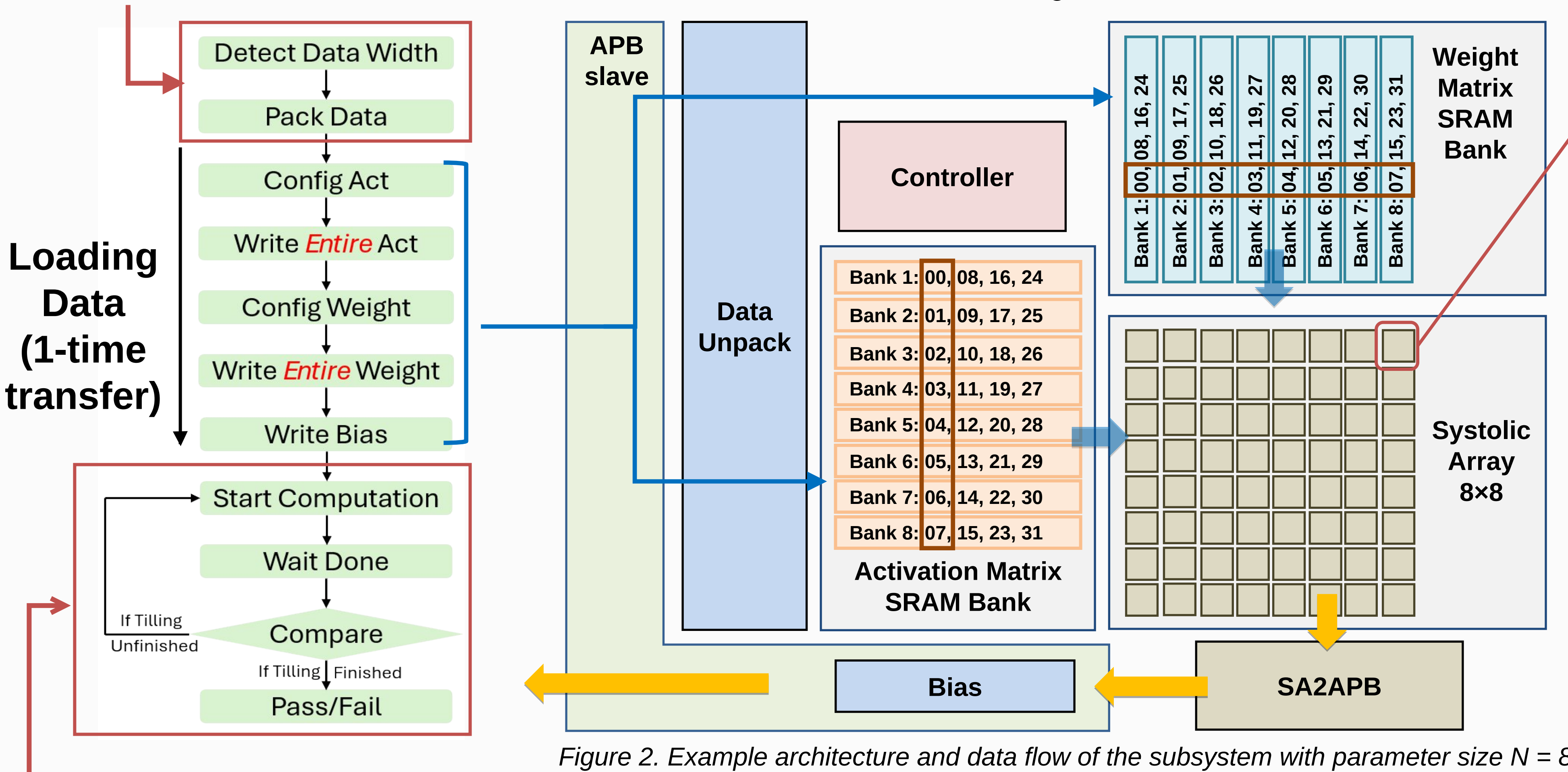


Figure 2. Example architecture and data flow of the subsystem with parameter size $N = 8$

Tiling Computation (Example of N = 8):

For each tile computation, 8 rows from the activation matrix and 8 columns from the weight matrix are taken and produce an 8×8 output tile. By iterating over different rows and columns, the Systolic Array is reused to generate all 16 tiles of the final 32×32 output matrix.

Support for different precisions and performance modes:

Precision: 4-bit to 32-bit dynamic support.

Two Modes: Combinational (High-speed) / Radix-4 Booth (Low-power & area).

Results for the Functional Design Part

Verification methods and procedures

- **Unit-level:** Apply direct stimuli to verify blocks in the subsystem.
- **Subsystem-level:** Use a UVM-based environment with coverage-driven stimuli.
- **SoC-level:** Perform co-simulation between a direct test-bench and C code tests.

FPGA Utilization (Implemented on Arty A7-100T Board)

	Slice LUTs	Slice Registers	BRAM Tiles	DSPs
Subsystem top	18,176	13,479	8	0
Systolic Array	17,181 (94.5%)	12,160 (90.2%)	0 (0.0%)	0 (0.0%)
APB – SRAM Wrapper	895 (4.9%)	1,246 (9.2%)	8 (100%)	0 (0.0%)
SA2APB	75 (0.4%)	40 (0.3%)	0 (0.0%)	0 (0.0%)
Controller	25 (0.2%)	33 (0.3%)	0 (0.0%)	0 (0.0%)

ASIC Synthesis (XFAB 180nm PDK)

Module	Area
Subsystem top (Total)	5,386,276 (100%)
APB – SRAM Wrapper	5,032,984 (93.4%)
4x4 Systolic Array	346,212 (6.4%)
SA2APB	4,443 (0.08%)
Controller	2,637 (0.05%)

- **Standard Cell:** 125,992 in total, including 53,894 combinational cells and 72,098 sequential/timing cells.
- **Dynamic Power:** ~59.4 mW @ 1.62V
- **Total Area Utilization:** ~240%
- **Timing check:** VIOLATED at the target operational frequency of 10 MHz. Worst slack is -10,604 ns.

Reason for high area utilization and timing violation: the PDK doesn't have SRAM hard-macro, so the SRAM RTL was synthesized to registers → Large area and timing violation caused by synthesized SRAM

SoC-level Behavioral Simulation result (f=100MHz)

Test	Input Matrices DW	Run-time (4x4 SA)	Run-time (8x8 SA)
0	32bit/32bit	1218.3us	901.5 us
1	4bit/4bit	743.1us	584.7 us
2	8bit/8bit	743.1us	663.9 us
3	16bit/16bit	901.5us	743.1 us
4	4bit/32bit	1139.1us	822.3 us
5	8bit/32bit	1139.1us	822.3 us
6	8bit/4bit	743.1us	584.7 us
7	16bit/32bit	1139.1us	822.3 us
8	16bit/4bit	901.5us	743.1 us
9	16bit/8bit	901.5us	743.1 us

Code Coverage (Subsystem-level Verification)

Metrics	Coverage	Metrics	Coverage
Branch	100%	Line	100%
Condition	100%	FSM	100%
Expression	100%	Toggle	100%

*Note: Some coverage results have been excluded correctly to achieve the above result.

Future Improvement

- Interrupt implementation
- Error detection mechanism
- Functional coverage verification
- Custom SRAM Standard Cell for better area utilization and timing